

Unit 1: Overview of Sterling File Gateway

This unit introduces you to the application terms and concepts of IBM® Sterling File Gateway. It also familiarizes you with the installation, user interfaces, setting up of users and trading partners in Sterling File Gateway. You will also learn to create and use routing channel templates, manage mailboxes and create static routes.

LESSON 1.1: Introduction

What this lesson is about

This lesson describes about the Sterling File Gateway installation, features, terms and concepts. It also introduces the new features and enhancements.

What you should be able to do

- Describe the major functions of Sterling File Gateway.
- Explain the terms and concepts of Sterling File Gateway.
- List the enhancements in Sterling File Gateway V2.2.6.
- Install and start the Sterling File Gateway application.

How you will check your progress

The progress of the course is analyzed based on the successful application of the hands-on exercises.

Additional References

Documentation

- Overview of Sterling File Gateway:
https://www-947.ibm.com/support/entry/portal/documentation_expanded_list/other_software/sterling_file_gateway?productContext=854280298
 - System requirements of Sterling File Gateway:
http://www-01.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_installation_226.doc/SFGSysRqmts.html
 - EBICS Integration:
http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_ebics.doc/SFGEB_IntegrationEBICS.html
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Introduction to Sterling File Gateway

Introduction

Sterling File Gateway is an application for transferring files between partners using different protocols, file name conventions, and file formats. Sterling File Gateway is useful for the movement of large and high-volume file transfers, with end-to-end visibility of file movement in a process-oriented and highly scalable framework that alleviates file transfer challenges, such as protocol and file brokering, automation, and data security.

Sterling File Gateway uses the IBM® Sterling B2B suite, which includes IBM® Sterling B2B Integrator, IBM® Sterling Standards, and the IBM® Sterling Services Oriented Architecture (SOA) platform to deliver capabilities similar to the ones found in IBM® Sterling Advanced File Transfer and IBM® Connect:Enterprise UNIX, while adding new features and functionality.

Sterling File Gateway can integrate with the IBM Sterling B2B Integrator EBICS server for secure file transfer between internal and external partners using same or different communication protocols.

EBICS

Electronic Banking Internet Communications Standard (EBICS) is a secure transport protocol and a B2B specification for electronic financial transactions between corporate firms and banks, and between banks.

It is an open standard, using state-of-the-art technology and the highest levels of security, such as:

- TCP-IP (Internet), HTTP, TLS (SSL), XML, ZIP, checkpoint-restart and so on.
- Encryption and authentication at both transport and application layers

EBICS continues to grow in acceptance across Europe, for many types of secure communication. EBICS communication has become Any-to-Any between participants, that include:

- Bank to Bank
- Corporate to Bank(s)
- Financial Services to Banks (Clearing Center)
- Insurance to Banks

Features

Sterling File Gateway provides many features:

- File and File name Transformations:
 - Flexible and powerful way to map input to output file names, using regular expressions.
 - Support for system-wide and group policies and partner-specific definitions.
 - Pre-built support for common file processing tasks.
 - Compression and decompression, format conversions, and including PGP encryption and decryption.

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Introduction to Sterling File Gateway

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Introduction

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Features

- Enhanced Visibility:
 - Events that are recorded for monitoring and reporting.
 - Detailed tracking for input/output file structure processing and dynamic route determination (runtime resolution decisions)
 - Ability to view all Sterling File Gateway data flows for all users.
- Partner Interface (myfilegateway):
 - Web browser-based interface that helps partners to upload and download files.
 - Subscribe to notifications of events, manage passwords, search.
 - View file transfer activity, and generate reports about file transfer activity.

Terms and Concepts

Sterling File Gateway operates in the file transfer space that has its own terms and concepts to describe its day to day operations. The following table provides some of the terms and their definitions:

Term	Definition
Consumer	Consumer partner who receives files directly or in a mailbox.
Consumer File Structure (CFS)	Description of consumer requirements for file naming and format structure.
Delivery Channel	Consumer side of the routing channel that specifies a consumer file structure and a delivery destination. There can be more than one delivery channel for each routing channel.
Event	Description of a distinct routing activity occurrence.
Fact	A named fragment of information that is gleaned from a file as it is identified and matched against routing channels. Facts can be derived from: ■ Part of a file name ■ Time message that is arrived from the transfer channel of the producer ■ Name of the producer ■ Name of the consumer ■ Automatically derived by the system
File Layer	Description of format. A file can encapsulate one or many other files. For example, a compressed file that contains a JPEG file. In this example, the compressed file is a container layer because it contains another file. The JPEG file is a non-container layer.

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Term	Definition
File Structure	Description of a basic content structure of the file and naming conventions. There are two types: ■ Consumer File Structure ■ Producer File Structure
Integration Architect	Type of user who creates, manages, and deletes partner groups, routing channel templates, producer file structures, and consumer file structures.
Mailbox	A repository for files with a hierarchical structure similar to files and directories on Windows and UNIX.
Notification	Email sent to a subscriber to notify that an event has occurred.
Mailbox Virtual Root	A position in the mailbox hierarchy that is associated with a user account and acts as the user's root directory.
Operator	Type of user who monitors system status and file activity. Troubleshoots problems with Partner activity, with the help of logs, reports, and notifications.
Partner User	Type of user who uploads and downloads files from myfilegateway portal. Views their own activity, specifies notifications to receive, and generates reports. Can be a producer, a consumer, or both.
Producer	Partner who creates and sends files.
Producer File Structure (PFS)	Description of producer requirements for file naming and structure.
Redeliver	The operation that an Operator performs on a delivery to reattempt that delivery. This operation can be performed if a delivery has already been attempted.
Regular Expression	An industry standard pattern-matching language. Used in Sterling File Gateway for matching file names and gathering facts that is related to file names.
Replay	The operation that an Operator performs on a file to cause the system to reprocess that file, as if it were sent again by the producer.
Route	A route is all of the information about a transfer of a file from a producer through to delivery to the consumer. Each file is associated with a route.
Route Details	Route details contain the details about a route, including the endpoint, consumer, producer, start and end times, any errors that occurred, and other details. Hyperlinks are provided to data flows, communication sessions, RCTs, business processes related to the route, and the payload.
Route Provisioner	Type of user who creates and manages partners, group membership, and routing channels.

Term	Definition
Routing Channel	Matches incoming producer files to consumers according to the requirements in the routing channel template, then transforms and sends them to the correct consumer in the format and name that is specified in the consumer file structure.
Routable Payload	A file that is eligible to match a routing channel. The top-level file (exactly the way the producer sent it) is always routable; if the top-level file is a compressed file, then the files contained within that compressed file are also routable. A single payload must not match multiple routing channels. However, because a compressed file and its constituents are distinct routable payloads, each is independently routed and can match different routing channels.
System Administrator	Installs and maintains system software. Administrator creates initial users, configures services, adapters, perimeter servers, certificates, and the database for sending and receiving files.

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Enhancements

The following enhancements are now available in Sterling File Gateway v2.2.6.1:

- Mass File Replay has been updated with a different icon for the original file to indicate file with a link to replay and also the original file contains a link to the replayed file.
 - Sterling File Gateway is now automatically installed when you install Sterling B2B Integrator V5.2.6 or later using IBM Installation Manager, and select the Sterling File Gateway option in the Installation Manager.
 - Sterling File Gateway allows integration with IBM Partner Engagement Manager integration to help manage the on boarding of trading partners.
 - Listening Producer API allows Sterling File Gateway to access a user defined server for to the producers to initiate start the transaction and poll for document through Sterling File Gateway routes.
 - EBICS Operators can search for general activity and view a file-centric information display that includes routes and deliveries that resulted from EBICS activity.
 - Access and protocol support for FTP, SFTP, Connect:Direct, myfilegateway for high availability and disaster recovery.
 - High availability and disaster recovery in B2B architecture for active deployments, application level command, payload across data centers.
 - Integration with IBM Partner Engagement Manager and an universal on boarding solution.
 - Improved business agility by secure and flexible integration with any partners, systems.
 - Improved operational efficiency, business performance and empowered data movement.
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Installing Sterling File Gateway

Overview

Sterling File Gateway is available with Sterling B2B Integrator and available with a patch. The Sterling File Gateway JAR file is located inside the following directory:

/Install_dir/packages/filegateway_version.jar

The package is installed using Install Services similar to a patch.

Upon successful installation, the following URL opens:

http://server:port/filegateway



Important

You must have a valid license of Sterling File Gateway before you install the installer jar files. If you do not have a valid license file, the installation completes with a message that the application has been deployed successfully. However, when Sterling File Gateway is started, a URL with */dashboard* opens. A valid Sterling File Gateway license can still be applied after the Sterling File Gateway installation.

Exercise 1.1.1: Set up of lab environment

Introduction

This exercise provides you with an overview of the lab environment and instructions to access the lab environment.

The exercise is a three step process:

1. Outline the set up of lab environment.
2. Access the Skytap image URL provided by your instructor.
3. Access the lab environment.

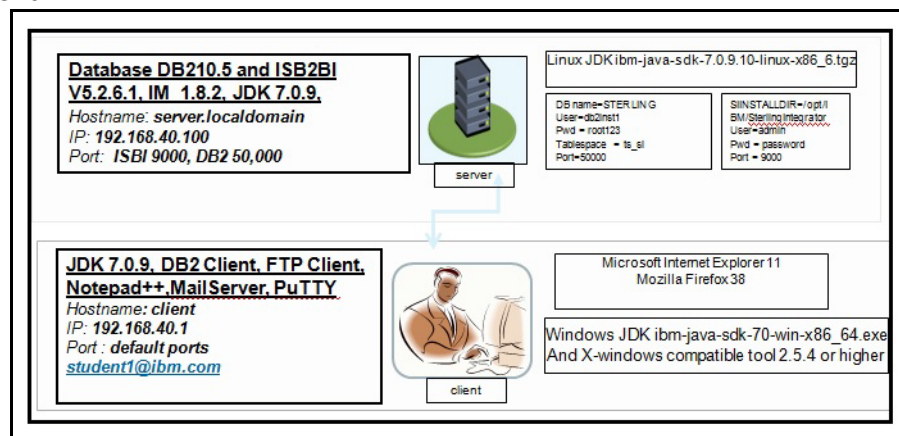
Instructions

Complete the following sets of procedure to access the lab environment and start the server.

Outline the set up of lab environment

The lab contains virtual images of two operating systems:

- Linux operating system, acting as a server, where database DB2, Sterling B2B Integration V5.2.6.1 are installed.
- Windows operating system acting as a client with respective tools such as PuTTY, Notepad++, DB2 client, Mail server, to access the server application. The screen here depicts the environment.



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Exercise 1.1.1: Set up of lab environment

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Instructions

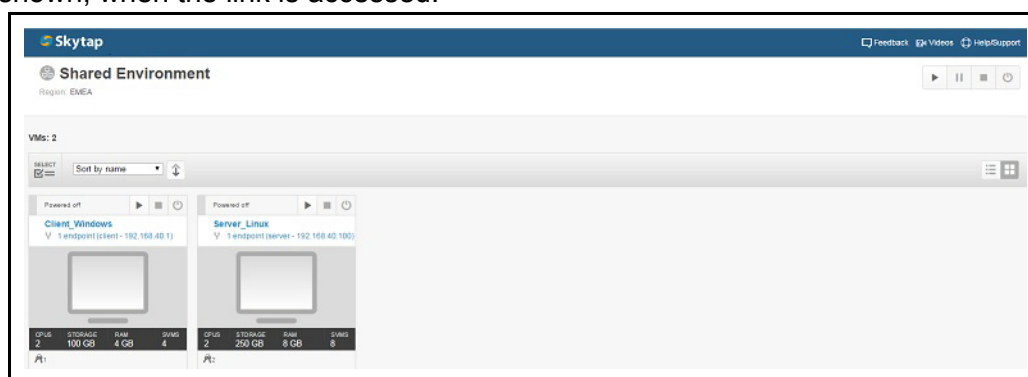
....(Continued)

Access the Skytap image URL provided by your instructor

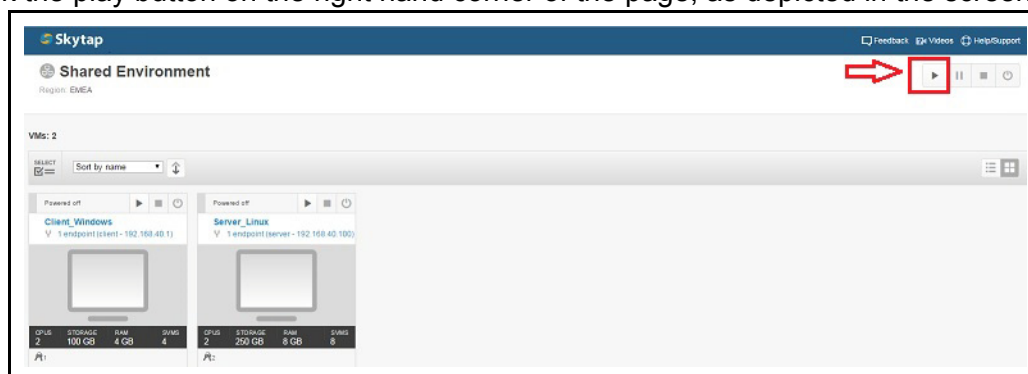
Each student is provided with a set of images to perform lab exercises. They are available through a cloud platform called Skytap. The instructor will provide a unique URL to each student to access the images. For example, the URL to access the image may resemble the following URL:

<https://cloud.skytap.com/vms/b759ebb4c4237519651ab8171c?????/desktop>

Step 1: In a web browser, open the URL you received from the instructor. You will see the screen shot as shown, when the link is accessed.



Step 2: Click the play button on image to start the machine. To start all the machines simultaneously, click the play button on the right hand corner of the page, as depicted in the screen shot..



Note

Ensure you suspend the image at the end of the training day. The controls to suspend, shut down, and power-off the images are available on the right hand corner of the URL page.

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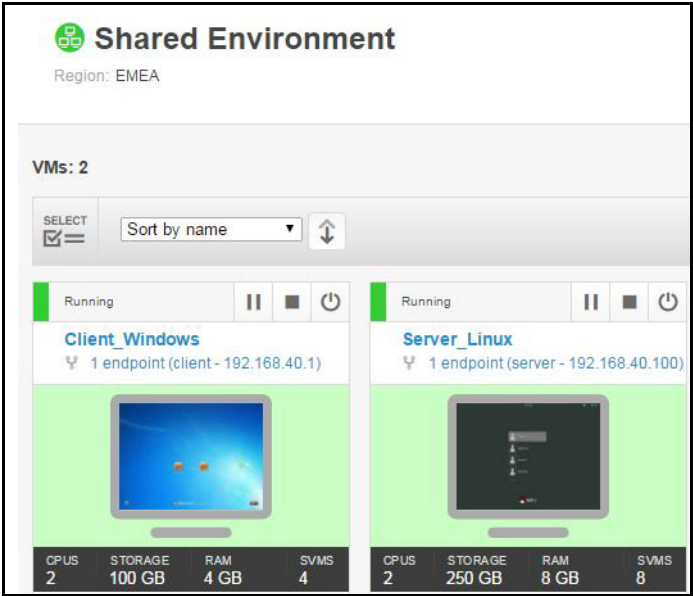
Exercise 1.1.1:Set up of lab environment

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Instructions

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Step 3: It may take a few minutes to start the image. You can view the progress of the image start up as displayed in the screen shot.



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Exercise 1.1.1: Set up of lab environment

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Instructions

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Access the lab environment

Step 1: Click windows image on the environment to access windows system.



Note

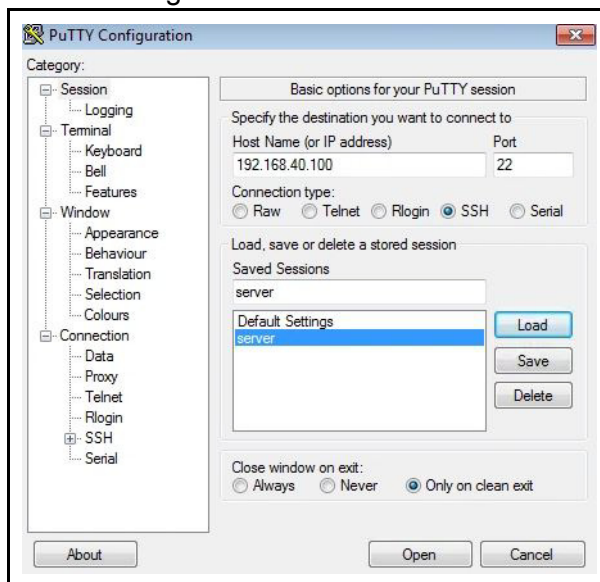
In the login window, specify **Administrator** as the user ID and **root123** as the password.

Step 2: Click **Enter**.

Step 3: From Desktop, click **PuTTY** exe file to access the server image.

Step 4: In the PuTTY Configuration window, select the **server** from the Saved Sessions list and click **Load**.

Step 5: Click **Open** to access server image.



Step 6: Click **No** in the PuTTY Security Alert window.

Step 7: Specify **root** as the user ID and root123 as the **password**, and click **Enter**.

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Procedure 1.1: Installing Sterling File Gateway

Introduction

The introduction, leads to know the Sterling File Gateway installation procedure. As a part of this course, Sterling File Gateway is completely set up. Sterling File Gateway is available with Sterling B2B Integrator installer. The Sterling File Gateway can be installed in one of the following ways:

- Installation Manager
- Install Service

Instructions

Installation manager

Sterling File Gateway is installed by selecting the Sterling File Gateway option in the installation manager during Sterling B2B Integrator V5.2.6.1 installation.

The instructions assume that you have the valid license to install Sterling File Gateway either on Linux or Windows operating system and have already installed the following software:

- JDK
- Database
- Installation manager

For detailed set of instructions to install Sterling File Gateway, refer to the

http://www.ibm.com/support/knowledgecenter/SS4TGX_2.2.0/com.ibm.help.sfg_installation_226.doc/sfg_installation_overview.html

The following steps provide you a high level overview of Sterling File Gateway installation.

Step 1: In the command prompt window, run the following export command to set the language to US English:

```
export LANG=en_US
```

Step 2: From the IBM Installation Manager directory, run the `./userinst` command to start the IBM Installation Manager.

Step 3: From left hand panel of the Installation Manager window, click **Install**.

Step 4: From the Install Package window, click the **repositories** link and add two repositories from the installer, for example

```
<Installer_folder>/common_pro/b2birepo/repository.conf  
<Installer_folder>/common_pro/gbmrepo/repository.conf
```

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Procedure 1.1: Installing Sterling File Gateway

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Instructions

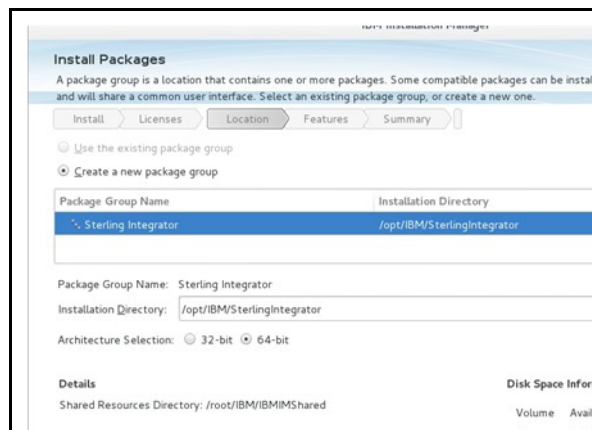
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Step 5: Click **Next**.

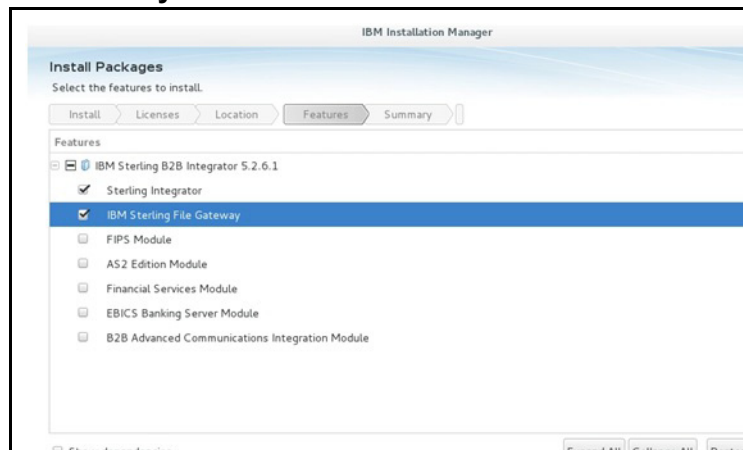
Step 6: Accept the license agreement.

Step 7: Click **Next**.

Step 8: In the Install Packages windows, specify the Sterling B2B Integrator installation directory, as shown in the screen shot:



Step 9: In the Install Packages window, navigate to the Features tab to select **Sterling Integrator** and **IBM Sterling File Gateway** and click **Next**.



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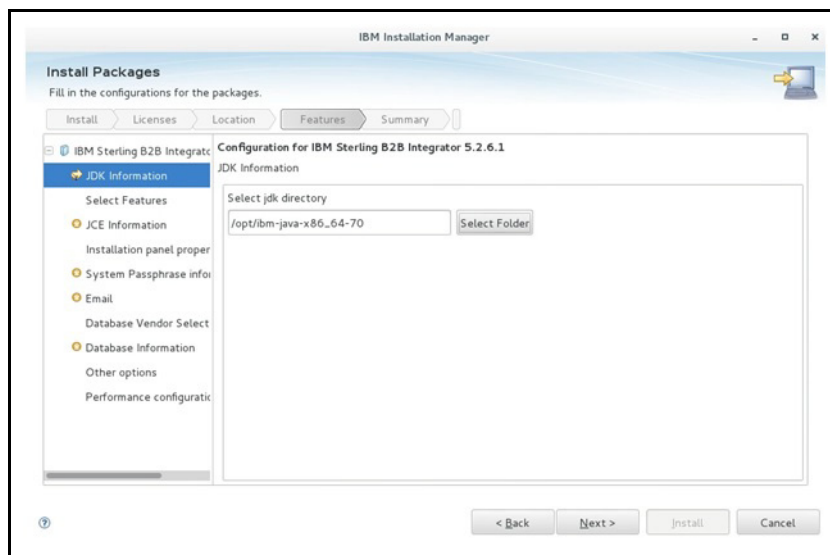
Procedure 1.1: Installing Sterling File Gateway

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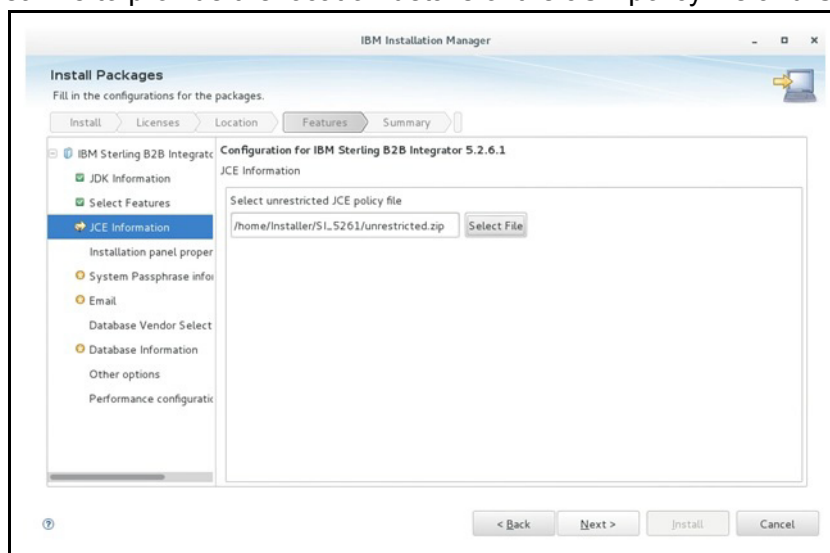
Instructions

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Step 10: In the configuration settings for Sterling B2B Integrator, click **Select Folder** to provide the path where JDK is installed and click **Next**.



Step 11: Click **Select File** to provide the location details of the JCE policy file and Click **Next**.



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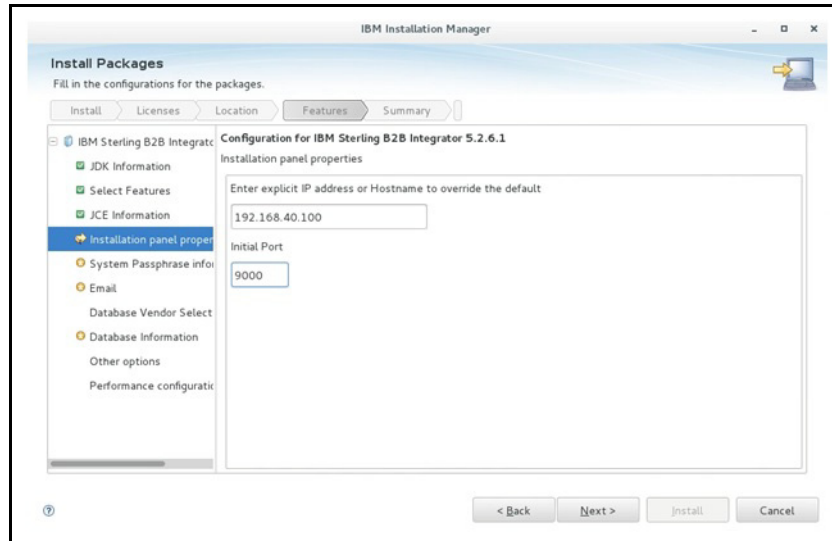
Procedure 1.1: Installing Sterling File Gateway

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Instructions

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Step 12: Specify the **hostname** or **IPaddress** and port that are a part of the installation panel properties and click **Next**.



Step 13: In the **System Passphrase info**, provide the **password**, as password, and click **Next**.

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Procedure 1.1: Installing Sterling File Gateway

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Instructions

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Step 14: Specify the e-mail address and the IP address of the **SMTP** mail server and click **Next**.

The screenshot shows the 'IBM Installation Manager' window with the 'Install Packages' tab selected. The left sidebar lists various configuration steps, with 'Email' highlighted. The main pane is titled 'Configuration for IBM Sterling B2B Integrator 5.2.6.1' and contains the 'Email' section. It has two input fields: 'E-Mail Address (in the format: you@yourdomain)' with the value 'student1@ibm.com' and 'SMTP Mail Server (IP Address or host name)' with the value '192.168.40.1'. At the bottom, there are buttons for '< Back', 'Next >', 'Install', and 'Cancel'.

Step 15: Specify the information related to the database in the lab environment, DB2 information is provided and click **Next**. The two screen shots provides the database details..

The screenshot shows the 'IBM Installation Manager' window with the 'Install Packages' tab selected. The left sidebar lists various configuration steps, with 'Database Information' highlighted. The main pane is titled 'Configuration for IBM Sterling B2B Integrator 5.2.6.1' and contains the 'Database Information' section. It has several input fields: 'Database user name' with the value 'db2inst1', 'Database password' and 'Database confirm password' both masked with '*****', 'Database catalog name' with the value 'STERLING', and 'Database host' with the value '192.168.40.100'. At the bottom, there are buttons for '< Back', 'Next >', 'Install', and 'Cancel'.

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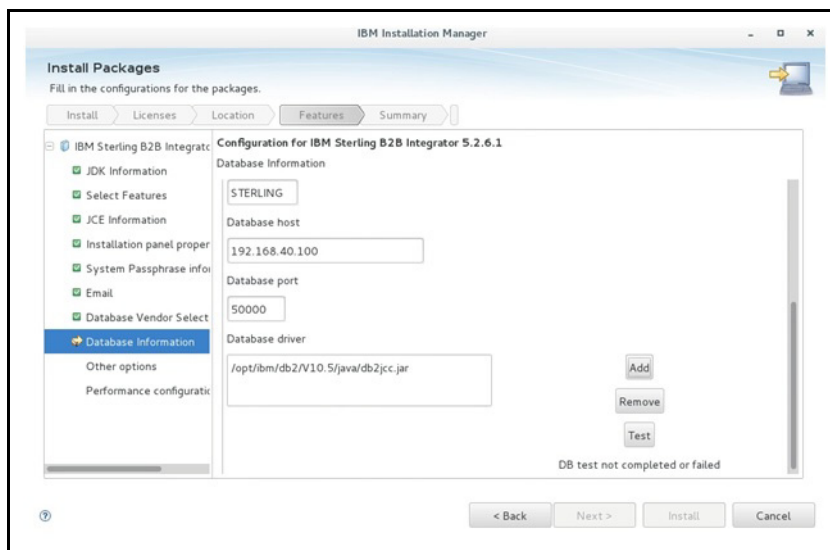
Procedure 1.1: Installing Sterling File Gateway

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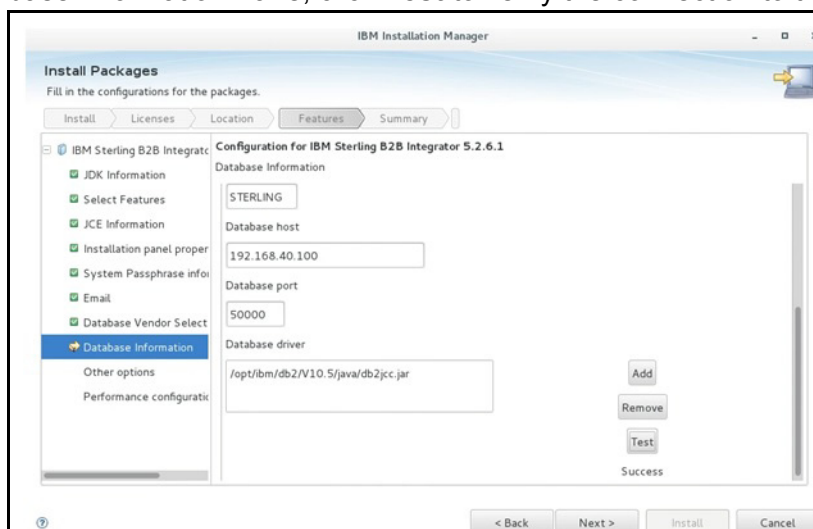
Instructions

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Step 16:



Step 17: In the Database Information Pane, click **Test** to verify the connection to the database.



Step 18: In the Other options panel, select **Verbose** install to view the detailed steps of the installation and click **Next**.

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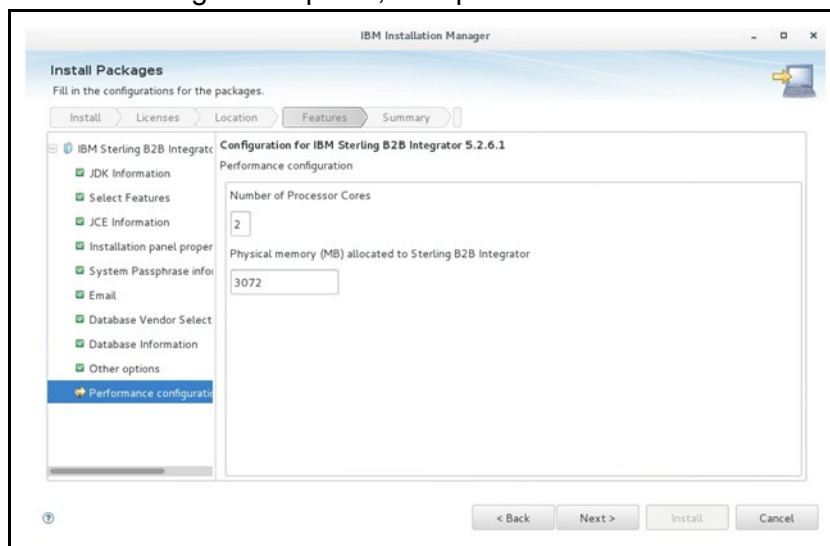
Procedure 1.1: Installing Sterling File Gateway

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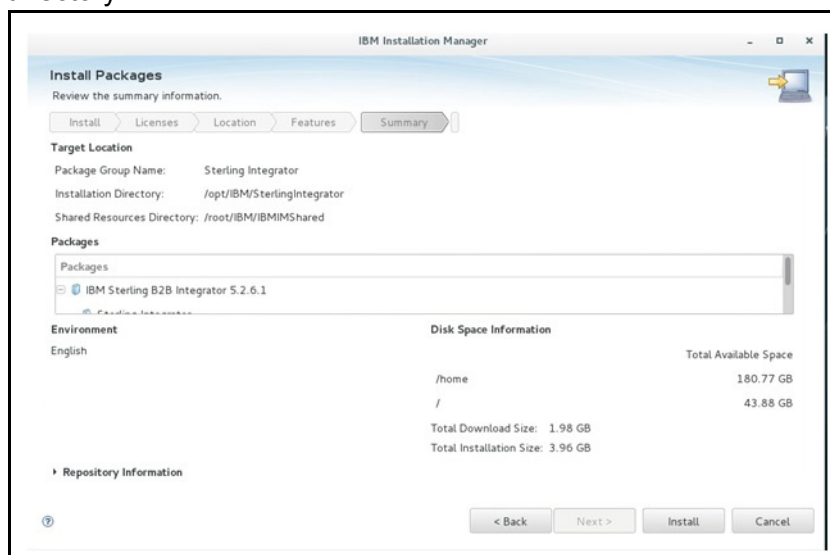
Instructions

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Step 19: In the Performance configuration panel, accept the default values and click **Next**..



Step 20: In the Summary tab, click **Install**, wait till installation manager completes the installation successfully. Any installation error, will be stored as *si.log* file, in Sterling B2B Integrator installation directory.



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Procedure 1.1: Installing Sterling File Gateway

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Instructions

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Important

It is strongly recommended to install Sterling File Gateway V2.2.6.1 during Sterling B2B Integrator V5.2.6.1 installation.

Install Service

The following set of instructions assumes that you have successfully installed Sterling B2B Integrator v5.2.6.1 and need to install Sterling File Gateway V2.2.6.

Complete the following steps to install Sterling File Gateway v2.2.6.1:

Step 1: Stop Sterling B2B Integrator using `./hardstop.sh` command from the install directory of sterling B2B Integrator; `<si_install_dir>/bin` and press **Enter**.



Note

If database password is encrypted, decrypt the password for the installation. In the lab environment, the password is not encrypted.

Step 2: Navigate to the `<si_install_dir>/packages` directory.

```
[root@server packages]# pwd
/opt/IBM/SterlingIntegrator/install/packages
[root@server packages]# ls
b2bi4meg_1000601.jar      preinstall.jar
b2bi4spe_1000601.jar      preinstall_sec.jar
filegateway_2020400.jar   ps_4060400.jar
filegateway_2020601.jar   ps_4060601.jar
GenSrvNtAdapter.jar       si_engine_patch_SI_5020601.jar
gis_xi_adapter_4060400.sda si_final_5020400.jar
GSNTClient.jar            SWIFTNet_MCFG
install_core_60170101.jar upgrade_5020400.jar
lib                       v_ebics_client_10000342.jar
mobile                   v_ebics_client_1000601.jar
[root@server packages]#
```

Step 3: Run the following command from `<SI_Install_dir>/bin` directory.

```
./InstallService.sh /<InstallDir>/packages/filegateway_2020601.jar
```

Step 4: press **Enter**, to view the files of Sterling File Gateway application is extracted.



Note

if database password is decrypted, re-encrypt the password.

Step 5: From the install directory of Sterling B2B Integrator; `<si_install_dir>/bin` start `./run.sh` command, and press **Enter**.

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Exercise 1.1.2: Starting Sterling File Gateway

Introduction

This exercise provides you the scripts to start and stop Sterling File Gateway server.

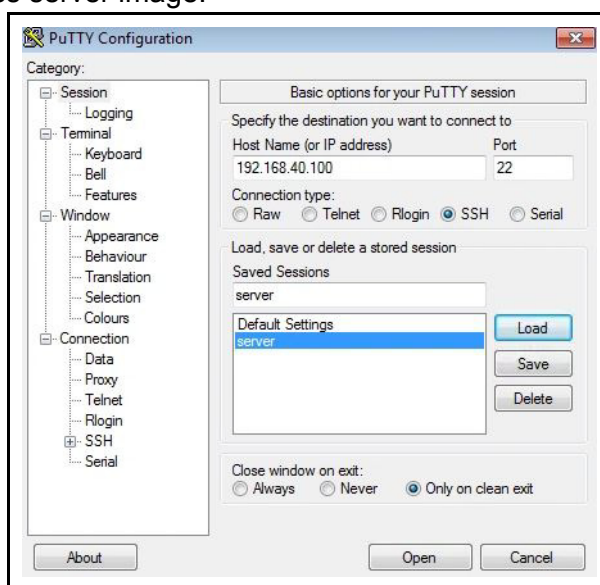
Instructions

Complete the following steps to stop the Sterling File Gateway server:

Step 1: From the windows image, access Desktop, double-click **PuTTY.exe** file.

Step 2: Select **server** and click **Load**.

Step 3: Click **Open** to access server image.



Step 4: Click **No** in PuTTY Security Alert window.

Step 5: Enter login as **root**, password as **root123**, and click **Enter**.

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Exercise 1.1.2: Starting Sterling File Gateway

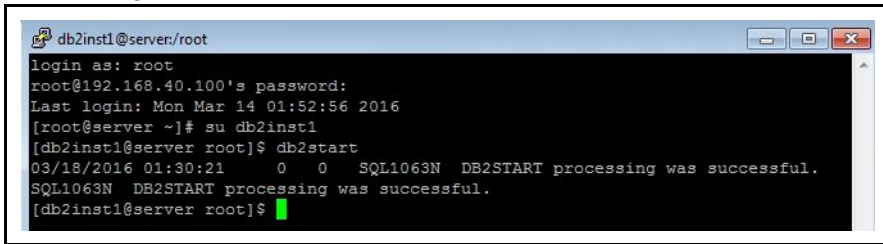
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Instructions

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Step 6: Login as database user to start DB2 database.

- i. From **PuTTY** window, enter `su db2inst1` and click **Enter**.
- ii. Enter `db2start`, and click **Enter**.
- iii. Upon successful start, the message '<db2start> processing was successful' is displayed, as shown in the image..



```
db2inst1@server/root
login as: root
root@192.168.40.100's password:
Last login: Mon Mar 14 01:52:56 2016
[root@server ~]# su db2inst1
[db2inst1@server root]$ db2start
03/18/2016 01:30:21    0    0    SQL1063N  DB2START processing was successful.
SQL1063N  DB2START processing was successful.
[db2inst1@server root]$
```

Step 7: Access Sterling B2B Integrator installation directory to start the Sterling File Gateway application.

Step 8: To switch user from **db2inst1** to root user, enter `su root` and click **Enter**.

Step 9: Enter password as **root123** and click **Enter**.

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Exercise 1.1.2: Starting Sterling File Gateway

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Step 10: Run the following command to change from existing directory location to the Sterling B2B Integrator installation bin directory. and press **Enter**

```
cd /opt/IBM/SterlingIntegrator/install/bin
```

Step 11: Enter `./hardstop.sh` command to stop Sterling File Gateway and press **Enter**.

Complete the following steps to start the Sterling File Gateway server:

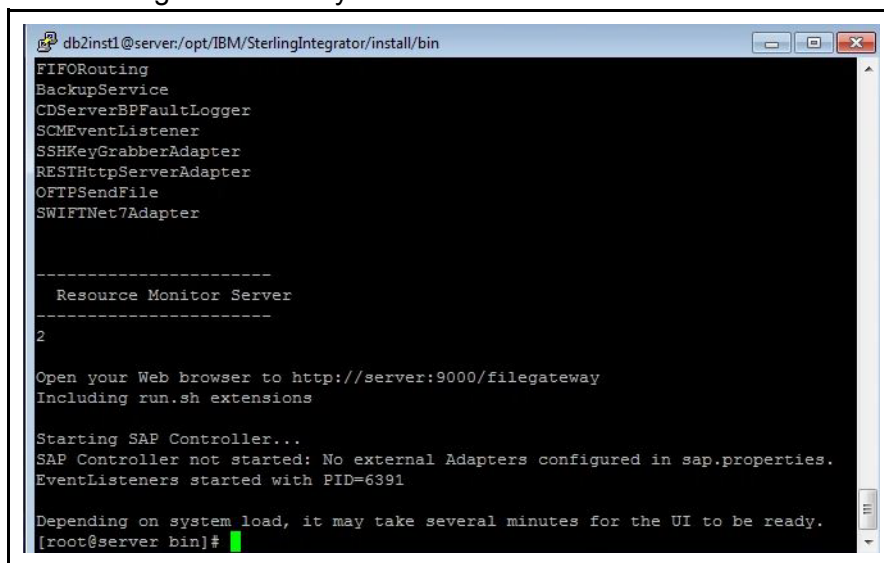
Step 1: From the windows image, access desktop.

Step 2: From **PuTTY** window, change to Sterling B2B Integrator installation bin directory using following command.

```
cd /opt/IBM/SterlingIntegrator/install/bin
```

Step 3: Enter `./run.sh` command to start the application.

Step 4: Enter **password** as the passphrase and press **Enter**. The screen shot displays a successful starting of the Sterling File Gateway Server.



```
db2inst1@server:/opt/IBM/SterlingIntegrator/install/bin
FIFORouting
BackupService
CDServeBPFaultLogger
SCMEventListener
SSHKeyGrabberAdapter
RESTHttpServerAdapter
OFTPSendFile
SWIFTNet7Adapter

-----
Resource Monitor Server
-----
2

Open your Web browser to http://server:9000/filegateway
Including run.sh extensions

Starting SAP Controller...
SAP Controller not started: No external Adapters configured in sap.properties.
EventListeners started with PID=6391

Depending on system load, it may take several minutes for the UI to be ready.
[root@server bin]#
```

Step 5: Open a web browser to, access `http://192.168.40.100:9000/filegateway` URL.

Step 6: To view the Sterling File Gateway system administrator user interface, specify **fg_sysadmin** as the user ID and **password** as the password.

Lesson review

What you have been able to do

- Describe the major functions of Sterling File Gateway.
 - Explain the terms and concepts of Sterling File Gateway.
 - List the enhancements in Sterling File Gateway V2.2.6.
 - Install and start the Sterling File Gateway application.
-